Project: Worked examples
Title: Limits and fits (ISO 286)
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1. Scope and method

Calculates the limits of size of a hole and a shaft for a nominal size and ISO 286 tolerance classes, and the resulting fit with its extreme clearances and interferences. It uses the ISO 286-1 formulae, not the tables.

WHAT IT CALCULATES	RELATION
Standard tolerance factor at the geometric mean D of the size step	i = 0.45·D^(1/3) + 0.001·D (µm)
Width of the tolerance band for the grade	ITn = multiplier × i
Fundamental deviations by letter from the published formulae, and hole deviations by the general rule	$EI_{hole} = -es_{shaft}$ (A–H); $ES_{hole} = -ei_{shaft} + \Delta$ (K–ZC)
Limits of size of the hole and the shaft	limit = nominal + deviation
Fit extremes, the largest and smallest clearance (negative is interference)	$c_{max} = ES_{hole} - ei_{shaft}$; $c_{min} = EI_{hole} - es_{shaft}$
Rolling-bearing bore mode	a bearing bore is not an ISO 286 hole; it has its own tolerance (ISO 492) lying below nominal, so ES_{hole} and EI_{hole} are entered from the bearing record instead of being computed

Basis: Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations[1](§4, §5); Tables of standard tolerance classes and limit deviations[2].

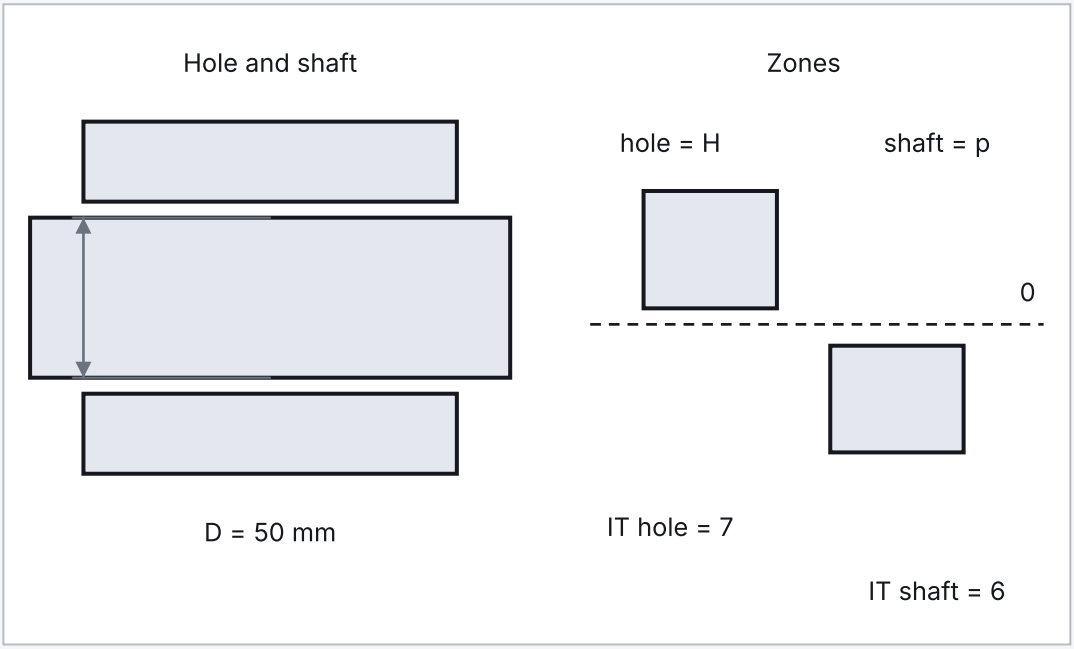


Figure 1 — A hole and its shaft of one nominal size D, and their tolerance zones about the zero line (hole above, shaft below as drawn; the letters place them). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$hole_{kind}$	Hole: an ISO 286 class, or a rolling-bearing bore (ES_{hole} and EI_{hole} from the bearing record, ISO 492)	ISO 286 hole class		default	any
$hole_{letter}$	Hole fundamental deviation letter	H		spec: DRG-HUB-04: H7/p6	any
$hole_{grade}$	Hole IT grade	7		spec: DRG-HUB-04: H7/p6	5 … 11
$shaft_{letter}$	Shaft fundamental deviation letter	p		spec: DRG-HUB-04: H7/p6	any
$shaft_{grade}$	Shaft IT grade	6		spec: DRG-HUB-04: H7/p6	4 … 11
D	Nominal size	50	mm	drawing: DRG-HUB-04 rev C	1 … 500 mm

3. Assumptions

A1. IT values and deviations are the unrounded results of the ISO 286-1 formulae at the size step's geometric mean. (The tables apply the standard's own rounding, so tabulated values differ by about 1 µm for the letters a to n.)
A2. For the letters p to zc the formula value can lie several µm below the tabulated deviation. (The tables add their own increments and sub-steps, so p6 at 450 mm is 62.2 µm from the formula against 68 µm tabulated.)
A3. Hole deviations follow the general rule from the shaft deviations, with Δ where the standard applies it. (Δ is ITn − IT(n−1), and the standard's table-specific departures are not reproduced.)

4. Working

[7] D_{step} = size step (lower, upper) [1](§4)
= (30, 50] mm
= 38.73 mm
Geometric mean of the size step

[8] i = 0.45·D^(1/3) + 0.001·D [1](§5)
= D = 38.73 mm
= 1.5612 µm
Standard tolerance unit

[9] IT_{hole} = IT7 [1](§5)
= 16 × i
= 24.98 µm
Hole tolerance

[10] IT_{shaft} = IT6 [1](§5)
= 10 × i
= 15.612 µm
Shaft tolerance

[11] es_{shaft} = ei_{shaft} + IT_{shaft} [1](§5)
= 24.98 + 15.612
= 40.592 µm
Shaft upper deviation

[12] ei_{shaft} = fundamental deviation 'p' [1](§5)
= +IT7
= 24.98 µm
Shaft lower deviation
[13] $ES_{hole} = EI_{hole} + IT_{hole}$ [1](§5)
= 0 + 24.98
= 24.98 µm
Hole upper deviation
[14] EI_{hole} = fundamental deviation 'H' [1](§5)
= $-es_{shaft}$ of the same letter
= 0 µm
Hole lower deviation
[15] $D_{hole,max} = D + ES_{hole}$
= 50 mm + 24.98 µm
= 50.025 mm
Hole maximum
[16] $D_{hole,min} = D + EI_{hole}$
= 50 mm + 0 µm
= 50 mm
Hole minimum
[17] $d_{shaft,max} = D + es_{shaft}$
= 50 mm + 40.592 µm
= 50.041 mm
Shaft maximum
[18] $d_{shaft,min} = D + ei_{shaft}$
= 50 mm + 24.98 µm
= 50.025 mm
Shaft minimum
[19] $c_{max} = ES_{hole} - ei_{shaft}$
= 24.98 µm − 24.98 µm
= 0 µm
Maximum clearance (negative: minimum interference)
[20] $c_{min} = EI_{hole} - es_{shaft}$
= 0 µm − 40.592 µm
= -40.592 µm
Minimum clearance (negative: maximum interference)
[21] $i_{max} = es_{shaft} - EI_{hole}$
= 40.592 µm − 0 µm
= 40.592 µm
Maximum interference
[22] $i_{min} = ei_{shaft} - ES_{hole}$
= 24.98 µm − 24.98 µm
= 0 µm
Minimum interference (negative: maximum clearance)
[23] fit = fit class
= H7/p6
= "interference"
Fit kind: clearance, transition or interference
[24] $c_{mean} = (c_{max} + c_{min}) / 2$
= (0 µm + -40.592 µm) / 2
= -20.296 µm
Mean clearance

5. Results

SYMBOL	VALUE	DESCRIPTION
c_{max}	0 µm	Maximum clearance
c_{min}	-40.59 µm	Minimum clearance
i_{max}	40.59 µm	Maximum interference
i_{min}	0 µm	Minimum interference
ES_{hole}	24.98 µm	Hole upper deviation
EI_{hole}	0 µm	Hole lower deviation
es_{shaft}	40.59 µm	Shaft upper deviation
ei_{shaft}	24.98 µm	Shaft lower deviation
IT_{hole}	24.98 µm	Hole tolerance
IT_{shaft}	15.61 µm	Shaft tolerance
$D_{hole,max}$	50.02 mm	Hole maximum
$D_{hole,min}$	50 mm	Hole minimum
$d_{shaft,max}$	50.04 mm	Shaft maximum
$d_{shaft,min}$	50.02 mm	Shaft minimum
fit	"interference"	Fit kind
c_{mean}	-20.30 µm	Mean clearance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fit classes against the fit kind	H7/p6	interference	–	

7. Applicability

WARNING $shaft_{letter}$: the deviation of 'p' is a formula value that can lie several µm below the table. At 450 mm, p6 gives 62.2 µm against 68 µm tabulated, so enter an override where the table governs. [interference-letter]

Defaults used (inputs left empty):
$hole_{kind}$: not entered, so the default of 'ISO 286 hole class' is used.

8. Verification

```
Template mech.iso286-fit@1.6.0 Content hash 296fa0d64befb299 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 1: Basis of Tolerances, Deviations and Fits. ISO 286-1:2010. ISO; 2010.
2. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 2: Tables of Standard Tolerance Classes and Limit Deviations for Holes and Shafts. ISO 286-2:2010. ISO; 2010.